



Regulatory analogues of nucleic acid synthesis screening



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Background

- Scientist Rob Carlson argues regulating distributed manufacturing technologies consistently backfires, and therefore regulatory requirements for nucleic acid synthesis screening may increase insecurity.
- He cites the printing press, Prohibition, methamphetamine, and 3D-printed firearms as cases where restrictions incentivised piracy, black markets, or undermining innovation.
- He challenges screening proponents to show how regulatory requirements will improve rather than worsen security.

Research objectives

- 1) Assess whether synthesis screening requirements may worsen the security landscape.
- 2) Identify regulatory regimes governing comparable products or technologies that can be judged successful.
- 3) Determine which ecosystem characteristics predict regulatory success or failure, and where synthesis screening sits relative to them.

Could regulations create greater insecurity?

- *New irresponsible entrants*: Most synthesis companies worldwide may already not screen so the baseline is already insecure; market dynamics like incumbent advantages and declining screening costs limit the viability of entrants competing commercially by merely not screening.
- *Provider relocation*: Overly stringent rules could push a major provider to an unregulated jurisdiction, but this argues against poor design, not regulation itself.
- *Underground synthesis market*: No pre-existing community seeks unscreened pathogenic sequences, making illicit synthesis economically unviable.
- *Evasion innovation*: Workarounds like oligonucleotide assembly exist but lack the profit-driven demand to improve reliability; most companies may not screen anyway, so innovative evasion is currently unnecessary.
- *Costs to defensive research*: Screening costs are negligible relative to total research spending; delay is a more serious concern but most orders don't trigger flags and many providers synthesise in parallel with review.

Responsibility-enabling innovation

- Innovation can support compliance, not just undermine regulation, e.g. faster/cheaper screening tools and streamlined KYC platforms.
- The synthesis industry has uniquely strong compliance incentives: a bioterrorism event involving synthesised DNA could be traced to a specific company, posing an existential reputational threat to the firm and wider industry.
- This contrasts with regimes like pseudoephedrine, where commercial sources are rarely identifiable and harms are diffuse.

Regulatory outcome assessments

- Of 20 supply-side regulatory regimes assessed, only EU/UK/Australian ammonium nitrate restrictions and credit card detection networks achieved durable success.
- CBRN regimes also show partial but improving effectiveness.
- The dominant pattern is displacement rather than elimination: restrictions incentivise shifts to different chemistries, jurisdictions, or methods.
- Drug precursor controls produce short-term disruption then adaptation; CBRN regimes prevent catastrophic events but can't stop determined state actors; digital information controls fail because digital goods are freely replicable.

Ammonium nitrate regulations

- The EU's restrictions on ammonium nitrate (AN), which limit civilian access to AN above 16% nitrogen content, have been largely successful: no confirmed standalone AN attacks in the EU since 2014.
- Displacement to triacetone triperoxide (TATP) based explosives arguably represents a net security gain — TATP devices are person-borne (limiting yield) and prone to premature detonation.
- AN and synthesis screening share similarities: both are supplier precursor controls with low malicious demand and meaningful substitute friction.
- Key differences in jurisdictional enforcement, screening complexity, and technology trajectory mean synthesis screening faces additional challenges.

Comparisons of characteristics

- *Distributed manufacturing*: Indicated by low capital/expertise requirements, high substitutability, declining costs, and many suppliers. Synthesis currently resembles centralised CBRN regimes but is trending toward greater distribution.
- *Product*: Nine regimes score similarly to synthesis on product attributes. Physical products are easier to control than digital goods, but cross-border movement of nucleic acids is easy.
- *Manufacturing*: Select Agents and Australia Group controls are the closest analogues; current barriers resemble stable CBRN regimes but the cost trajectory mirrors failed digital regimes.
- *Buyer*: CBRN regimes are the closest match. Low illicit demand and downstream manufacturing requirements mean no profit motive for illicit supply chains and multiple attack-chain failure points persist.
- *Manufacturer/supplier*: CBRN and credit card fraud networks score highest. High reputational risk and self-regulation capacity are favourable; jurisdictional spread (600+ companies of unknown screening status) is the most concerning vulnerability.
- *Overall*: CBRN regimes are the most structurally similar to nucleic acid synthesis; achieving partial but sustained effectiveness. Synthesis diverges on technology trajectory, suggesting a time-limited window of effectiveness without adaptation.